

Assignment 1

Q1) Python was developed by Guido van Rossum in 1989. It was found to address the issues in ABC language.

Q2) Python is a high level language & C is a low level programming language.

Q2) 1) source code

2) Lexical analysis -

Here the python interpreter scans the code and breaks it into tokens.

3) Parsing & AST generation:

Tokens are passed & transformed into an AST. Basically it checks for syntax errors.

(4) compiled to Bytecode.

Bytecode is platform independent & stored in pycache folder.

(5) execution by PVM.

The pvm reads the bytecode & executes it byte by byte.

3) Abstract syntax tree (AST) is compiled into the byte code -

The byte code is platform independent & is stored in pycache folder.

The pvm reads the byte code.

& executes it line by line.

- 4) Python is an interpreted language. A python code is interpreted language which is executed line by line using an interpreter.
- 5) The bytecode of a python program is stored into the `-pycache-` folder.
- 6) In python we don't declare the datatype of a variable. It is decided at the time of runtime.

For eg :-

```
No1 = 10.  
print( No1 )  
print( type( No1 ) )
```

- 7) Python program can run on
- Windows
 - Linux
 - MacOS without changing the code.
- This makes python portable across different

- 8) Indentation means specifying a block in python language. A particular function is ~~not~~ developed after putting ':' and then giving an indentation just like we specify a block in C, C++ & Java.

9) If indentation is not given properly it will give an error as the code won't enter the function

```
def def func():  
    print("I am here")  
this.func()
```